

A · ONE-TIME SETUP — YOU, ONCE EVER

1 Create the signing key

`keytool -genkeypair → hcrobotics-release.jks`  
Valid 30 years. **This key can never change.**

▼

2 Save it locally

`keystore.properties` in the repo root.  
Gitignored — never committed.

▼

3 Add 4 GitHub Secrets

Settings → Secrets → Actions  
`KEYSTORE_BASE64` (the .jks, base64)  
`KEYSTORE_PASSWORD`  
`KEY_ALIAS` = `hcrobotics`  
`KEY_PASSWORD`

▼

4 Token permissions

Contents: Read+Write  
Workflows: Read+Write

▼

5 Install app once on device

Cable, or download the APK from the Releases page in the phone browser.

▼

6 Grant 2 permissions on phone

Notifications → ON  
Install unknown apps → Allow  
Without these the update fails **silently**.

▼

Setup complete — never needed again

B · MAKE A RELEASE — EVERY CHANGE

7 Branch from master

`git checkout -b release/vX.Y.Z master`

▼

8 Write the code

Feature or fix, in the app or a module.

▼

9 Bump the version

`app/build.gradle`  
`versionCode` 10 → 11 (must increase)  
`versionName` "1.9.0" → "2.0.0"

▼

10 Build & check locally

`gradlew :app:assembleRelease`  
`gradlew :app:lintRelease`

▼

11 Push the branch

`git push -u origin release/vX.Y.Z`

▼

12 Open a Pull Request

base `master` ← head `release/vX.Y.Z`

▼

13 Merge the PR

This is the trigger. Everything after it is automatic.

C · CI PUBLISHES — AUTOMATIC, ~2 MIN

14 Workflow starts

`.github/workflows/publish-ota-update.yml`  
on push to `master`

▼

15 Read version

Parses `versionCode` / `versionName` from `app/build.gradle`.

▼

16 new code > published code?

NO → stop quietly, nothing published

▼ YES

17 Restore the keystore

Decodes `KEYSTORE_BASE64` to a temp file.  
Missing secret → **build fails on purpose**.

▼

18 Build signed release APK

`assembleRelease` + R8 shrink.  
Same key every time. ~2 MB.

▼

19 Verify signature & hash

`apksigner verify`, then SHA-256 of the APK.

▼

20 Create Release + tag

Tag `vX.Y.Z`, APK uploaded as the release asset.

▼

21 Write the manifest

`ota/update-manifest.json`  
`version`, `apkUrl`, `sha256`, `size`, `notes`,  
+ `previous` = rollback target

▼

22 Commit manifest to master

This starts the rollout. The Release alone is invisible to devices.

D · ON THE PHONE — AUTOMATIC

23 App checks

On every app open — instant.  
Plus every ~6 h in background (WorkManager).

▼

24 Fetch the manifest

HTTPS GET of `raw.githubusercontent.com/.../ota/update-manifest.json`

▼

25 manifest code > installed code?

NO → "up to date", stop

▼ YES

26 Notification appears

"Update available — Version X.Y.Z"  
Also: dot on the Settings icon.

▼

27 User taps → dialog

Shows old → new version, download size, what's new.  
"Not now" keeps it in Settings.

▼

28 Download the APK

From the GitHub Release, into app-private storage.

▼

29 SHA-256 matches manifest?

NO → delete file, refuse to install

▼ YES

30 System install prompt

`PackageInstaller` → Android asks the user to confirm. Signature must match.

▼

31 Updated — app restarts

LIMITATIONS & TRAPS — read before releasing

- ! **The signing key can never change.** A different key = `INSTALL_FAILED_UPDATE_INCOMPATIBLE`. Lose the `.jks` and no device can ever be updated again — each must be uninstalled and reinstalled by hand. **Keep 2 backups.**
- ! **Debug and release are different apps.** Different applicationId and different key. Moving a device from debug to release needs one uninstall.
- ! **Forgetting to bump `versionCode`** means CI publishes nothing. No error, just no update.
- ! **CDN cache ~5 minutes.** A device checking immediately after publish may still receive the old manifest. Wait, then re-open.
- ! **Background check is "at least 6 h", not exactly.** Android batches it. Opening the app checks instantly.
- ! **Notifications off (Android 13+)** = update alerts silently dropped. Settings screen warns when this is the case.
- ! **"Install unknown apps" not granted** = install cannot start.
- ! **Rollback only works on debuggable builds.** Android blocks installing an older version over a newer one; on release builds the app must be uninstalled first (data is erased).
- ! **Repo must be public** for devices to download without a token. A token embedded in an APK is readable by anyone.
- ! **Never hand-edit `ota/update-manifest.json`.** CI owns it on master. Editing it on a branch causes merge conflicts — always keep master's copy.

TIMING — merge to phone

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| CI build + publish | ~2 min                                                                                  |
| CDN cache clears   | ~5 min                                                                                  |
| Phone, app opened  | seconds                                                                                 |
| Phone, left idle   | up to ~6 h                                                                              |
| Realistic total    | ~7 min if someone opens the app                                                         |
| Force it now       | Settings → Check for updates                                                            |
| Make it instant    | Add Firebase Cloud Messaging → call <code>OtaUpdater.checkNow()</code> (free, optional) |

WHAT IS CONFIGURED WHERE

|                   |                                                                      |
|-------------------|----------------------------------------------------------------------|
| Manifest URL      | <code>AppConfig.UPDATE_MANIFEST_URL</code> — points at <b>master</b> |
| Check interval    | <code>AppConfig.UPDATE_CHECK_INTERVAL_HOURS</code>                   |
| Version           | <code>app/build.gradle</code>                                        |
| Signing (local)   | <code>keystore.properties</code> (gitignored)                        |
| Signing (CI)      | 4 GitHub Secrets                                                     |
| Pipeline          | <code>.github/workflows/publish-ota-update.yml</code>                |
| What devices read | <code>ota/update-manifest.json</code> (written by CI only)           |
| Update client     | <code>:updater</code> module — reusable in any app                   |
| Logs on device    | <code>adb logcat -s HCota</code>                                     |